Tianye (Jerry) Ding

Department of Computer Science, The University of Texas at Austin
Austin, TX | jerrydty@cs.utexas.edu
Homepage | Google Scholar | GitHub | LinkedIn | X

RESEARCH INTERESTS

Computer vision and geometric deep learning, with an emphasis on bridging interpretable classical formulations and modern learned models. My work studies how explicit structural priors—geometry, compositionality, and physical laws—unlock capabilities that scale alone does not, spanning efficient and mathematically grounded perception algorithms as well as 3D and 4D generative models that remain physically consistent across space and time.

EDUCATION

Ph.D., Computer Science 08/2026 – Present
The University of Texas at Austin
Co-advised by Georgios Pavlakos and Qixing Huang.

B.S., Computer Science, *magna cum laude* 09/2020 – 04/2024
Khoury College of Computer Sciences, Northeastern University

PUBLICATIONS

* denotes equal contribution; † denotes equal advising.

Conference Papers

UniCorrn: Unified Correspondence Transformer Across 2D and 3D
Prajnan Goswami*, **Tianye Ding***, Feng Liu, Huaizu Jiang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
arXiv | Project Page | Code

LASER: Layer-wise Scale Alignment for Training-Free Streaming 4D Reconstruction
Tianye Ding*, Yiming Xie*, Yiqing Liang*, Moitreya Chatterjee, Pedro Miraldo, Huaizu Jiang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
arXiv | Project Page | Code

Struct2D: A Perception-Guided Framework for Spatial Reasoning in Large Multimodal Models
Fangrui Zhu*, Hanhui Wang*, Yiming Xie, Jing Gu, **Tianye Ding**, Jianwei Yang, Huaizu Jiang
Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
arXiv

ODTFormer: Efficient Obstacle Detection and Tracking with Stereo Cameras Based on Transformer
Tianye Ding*, Hongyu Li*, Huaizu Jiang
IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2024.
Selected for oral pitch and interactive presentation.
arXiv | Project Page | Code

RESEARCH EXPERIENCE

Graduate Research Assistant 08/2026 – Present
The University of Texas at Austin, Austin, TX
Co-advised by Georgios Pavlakos and Qixing Huang.

Research Assistant 01/2023 – 06/2026
Visual Intelligence Lab, Northeastern University, Boston, MA
Computer vision research under the supervision of Huaizu Jiang. Led work on unified 2D–3D correspon-

dence estimation, training-free streaming 4D reconstruction, and efficient stereo-based obstacle detection and tracking.

Machine Learning Research Engineer Intern 08/2022 – 12/2022
Deep Ivy Inc.
Developed and maintained Ivy, an open-source unified machine learning framework.
Repository: `github.com/unifyai/ivy`

TALKS AND PRESENTATIONS

ODTFormer: Efficient Obstacle Detection and Tracking with Stereo Cameras Based on Transformer
Tianye Ding, Hongyu Li, Huaizu Jiang
Spotlight presentation, IROS 2024 Workshop on Planning, Perception and Navigation for Intelligent Vehicles, Abu Dhabi, UAE, October 2024.

AWARDS AND HONORS

Magna Cum Laude, Northeastern University 04/2024

Dean's List, Northeastern University 2021 – 2023

TEACHING EXPERIENCE

CS 2510: Fundamentals of Computer Science II 01/2022 – 05/2022
Teaching Assistant, Khoury College of Computer Sciences, Northeastern University

SELECTED WRITING

The Underrated Decoder Behind a Unified Correspondence Model, May 2026.
Why Cost Volume Construction Can Be a Non-Trivial Yet Interesting Problem in Transformer-Based Models?, October 2024.

TECHNICAL SKILLS

Languages: Python | C++ | C | Java | MySQL | LaTeX
Frameworks and Libraries: PyTorch | NumPy | TensorFlow | OpenCV | OpenGL
Tools and Systems: Git | Slurm | Linux | macOS | Windows | JetBrains Suite | Adobe Creative Suite